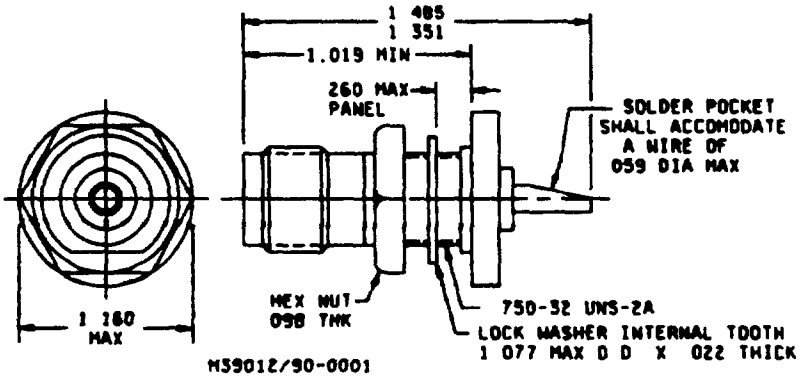
NOTE The document identifier and heading has been changed on this page to reflect that this is a performance specification There are no other changes to this document The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised

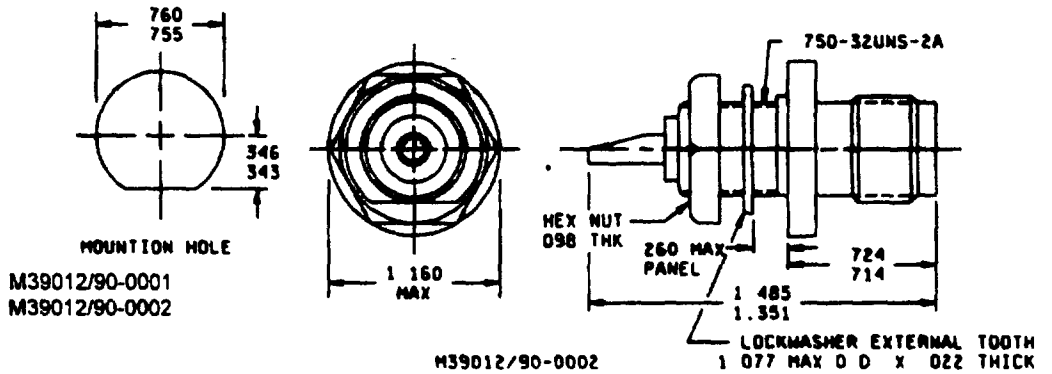
PERFORMANCE SPECIFICATION

CONNECTORS, COAXIAL, RADIO FREQUENCY, SERIES QNC (UNCABLED - RECEPTACLE, SOCKET CONTACT, JAM NUT, FRONT AND REAR MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-PRF-39012



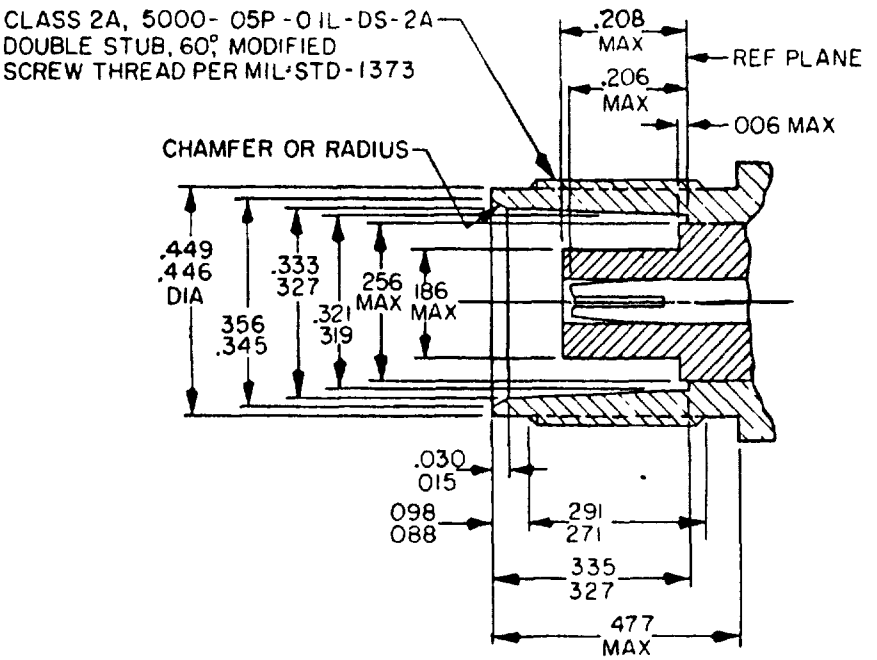


INCHES	MM
022	56
062	1 57
098	2 50
260	6 60
343	8 71
346	8 79
714	18 14
724	18 39
750	19 05
755	19 18
760	19 30
1 077	27 36
1 091	27 71
1 160	29 46
1 351	34 32
1 485	37 72

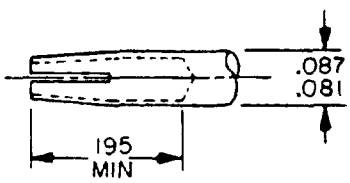
NOTES
1. Dimensions are in inches
2. Dimension 1 160 is the largest overall diameter
3. Dimension 1 485 is the largest overall length
4. Metric equivalents (to the nearest 01 mm) are given for general information only and are based upon 1 inch = 25 4 mm
5. Wrench flats to accommodate standard wrench opening per H-28, app 10

FIGURE 1 General configuration

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INCHES	MM	INCHES	MM	INCHES	MM
.006	.15	.186	4 72	.327	8 31
.015	.38	.195	4 95	333	8 46
.030	76	.206	5 23	335	8 51
.052	1.32	208	5 28	345	8 76
.054	1.37	.256	6 50	356	9 06
.081	2.06	.271	6.88	.446	11 32
.087	2.21	.291	7.39	449	11 40
.088	2.24	.319	8 10	477	12.12
.098	2.50	.321	8 15	5000	12.70



NOTES:
1. Dimensions are in inches
2. Metric equivalents (to the nearest 01 mm) are given for general information only and are based upon 1 inch = 25 4 mm
3. The .256 dimension applies to that portion (if applicable) of the dielectric which protrudes beyond the metal shoulder (or reference plane) by dimension .006.

FIGURE 2. Mating dimensions for socket terminations.

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ENGINEERING DATA

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating: 500 volts rms maximum working voltage. 125 volts rms maximum at 70,000 feet.

Temperature rating: -65° to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:
Longitudinal force - Not applicable.
Torque - 2 inch-pounds maximum.

Coupling proof torque:
20 inch-pounds minimum.

Inspection conditions: 4 to 6 inch-pounds.

Mating characteristics:
See figure 2 for dimensions.
Center contact (socket):
Oversize test pin - .057 diameter minimum (nonclosed entry contacts only).
Insertion depth - .125 minimum.
Number of insertions - One.

Insertion force test - Steel test pin dia .054 minimum.
Test pin finish - 16 microinches.
Insertion force - 2 pounds maximum.

Withdrawal force test: Steel test pin dia .052 maximum.
Withdrawal force - 2 ounces minimum.
Test pin finish - 16 microinches.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302, test condition B, MIL-STD-202. 5,000 megohms minimum.

Center contact retention: 6 pounds minimum.

Corrosion (salt spray): Method 101, test condition B, MIL-STD-202.

Voltage standing wave ratio (VSWR): Not applicable.

Connector durability:
500 cycles minimum at 12 cycles/min maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

	Initial	After environment
Center contact	1.5	2.0
Outer contact	.2	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 2,000 volts rms at sea level.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202.

Shock: Method 213, test condition I, MIL-STD-202.

Temperature cycling: Method 102, test condition C, MIL-STD-202.

Thermal shock: Not applicable.

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level: Not applicable.

RF high potential withstanding voltage:
Voltage and frequency - 1,000 volts rms at 5 MHz.
Leakage current - Not applicable.

Cable retention force: Not applicable.

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Coupling mechanism retention force: Not applicable.

RF leakage: Not applicable.

Insertion loss: Not applicable.

Qualification not required.

Part number: M39012/90 -0001 for the front- and M39012/90 -0002 for the rear-mounted connector.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU, MI, EL
DSA - ES

User activities:
Army - AT, AV, ME

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5935-1862-8)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-C-39012/90 CONNECTORS, COAXIAL, RADIO FREQUENCY, SERIES, QNC (UNCABLED-RECEPTACLE, SOCKET CONTACT, JAM NUT, FRONT AND REAR MOUNTED

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

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U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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